

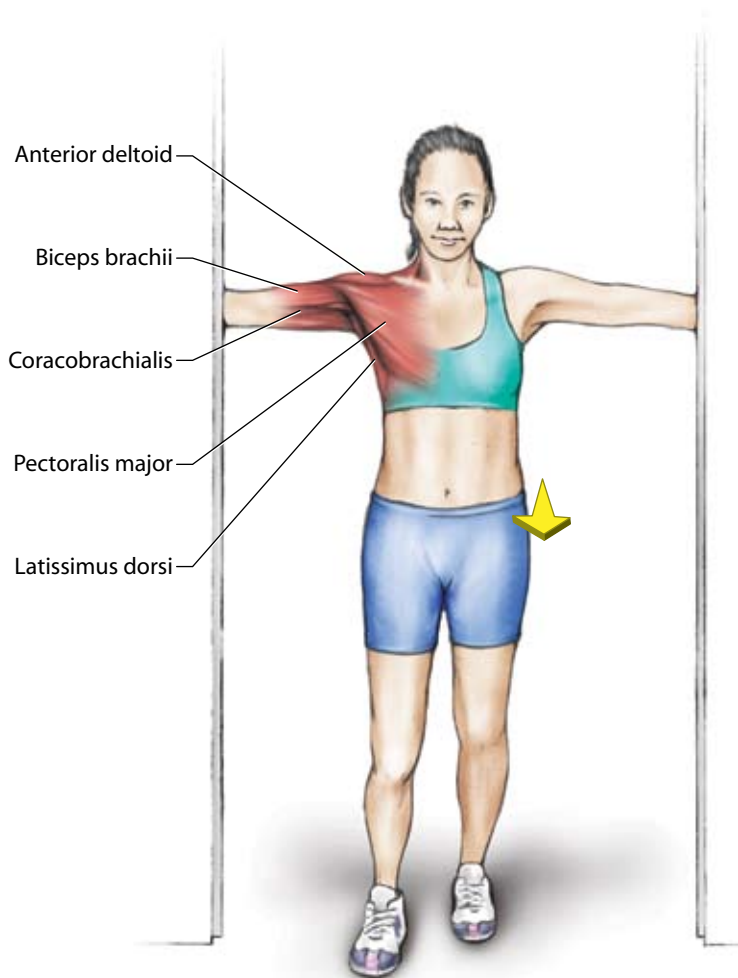
SHOULDERS, BACK, AND CHEST

There are five major pairs of movements at the shoulder: flexion and extension, abduction and adduction, external and internal rotation, retraction and protraction, and elevation and depression. Illustrations showing these movements as well as a chart showing which muscles do specific movements are located at the end of the chapter (pages 21-23). The bones of the shoulder joint consist of the humerus (upper arm bone), scapula (shoulder blade), and clavicle (collarbone). The scapula and clavicle essentially “float” on top of the rib cage. Therefore, a major function of many upper-back and chest muscles is to attach the scapula (in the upper back) and clavicle (in the upper chest) to the rib cage and spine. This provides a stable platform for arm and shoulder movements. Of the five movement pairs mentioned here, retraction and protraction and elevation and depression are usually classified as stabilization actions. The majority of the muscles involved in moving and stabilizing the shoulder bones are located posterior. The scapula is a much larger bone than the clavicle and has room for more muscles to attach. The posterior muscles are infraspinatus, latissimus dorsi, levator scapulae, rhomboids, subclavius, subscapularis, supraspinatus, teres major, teres minor, and trapezius (attached to the upper posterior rib cage, vertebrae [spine], and scapulae), as well as the deltoid and triceps brachii (attached to the scapulae and humerus). The anterior muscles are the pectoralis major (attached to the clavicle anterior rib cage and humerus), the pectoralis minor, and serratus anterior (attached to the anterior rib cage and anterior scapulae as well as the biceps brachii, coracobrachialis, and deltoid (attached to the anterior scapulae and humerus).

Common complaints associated with the musculature of the shoulders and upper back and chest are tight muscles and muscle spasms in the neck (middle and upper trapezius), shoulder (trapezius, deltoid, supraspinatus), and upper-back muscles (rhomboids and levator scapulae). Interestingly, the tightness felt in these muscles is usually a result of initial tightness in their antagonist muscles. In other words, tight muscles in the upper chest caused the tightness felt in the upper back. Tight chest muscles (that is, the pectoralis major) cause a constant low-level stretch on the muscles of the upper back. Eventually, this low-level stretch elongates the ligaments and tendons associated with the upper-back muscles. Once these ligaments and tendons become elongated, the tone in their associated muscles falls dramatically. To reclaim the lost tone, the muscles must increase their force of contraction. Increased force in turn causes more stretch of the ligaments and tendons, and increased muscle contraction must compensate for that. Hence, a vicious cycle commences. The best way to prevent or stop this cycle is to stretch the anterior shoulder and chest muscles. As these muscles increase in flexibility, the tightness of the posterior muscles will also be reduced. Also, immediately after stretching, the strength of the muscles is diminished. It is also a good idea to stretch the opposing muscles just before and immediately after working any group of muscles. If this is done three or more times a week, the muscles will actually increase in flexibility and gain strength. Stretching will also reduce the frequency of tightness for any group of muscles.

Many of the instructions and illustrations in this chapter are given for the left or right side of the body. Similar but opposite procedures would be used for the opposite (not pictured) side of the body.

Shoulder Flexor Stretch



Technique

Stand upright while facing a doorway or corner.

Place feet shoulder-width apart with one foot slightly in front of the other.

With straight arms, raise your arms to shoulder level and place the palms on the walls or doorframe with the thumbs on top.

Lean the entire body forward.

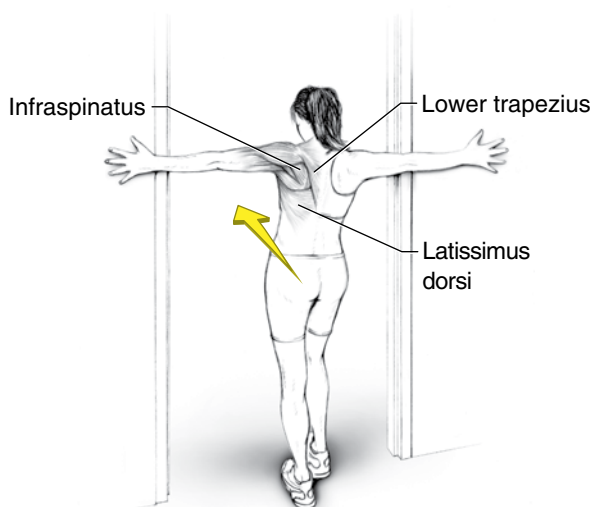
Muscles Stretched

Most-stretched muscles: Pectoralis major, anterior deltoid, coracobrachialis, biceps brachii.

Lesser-stretched muscles: Infraspinatus, latissimus dorsi, subclavius, lower trapezius.

Commentary

To get the maximum benefit during the stretch, keep the elbows locked and the spine straight. The greater the forward lean, the better the stretch. Forward lean is controlled by how far the lead foot is in front of the chest at the start position. Hence, place the foot forward only enough to maintain balance. It is possible to do the neck extensor stretch simultaneously with the shoulder flexor stretch. However, without having the hands pushing down on the head, the neck extensor stretch will be of a lower intensity than if it were done by itself.



VARIATION

Shoulder Flexor and Depressor Stretch

Elevating the arms will stretch more muscles.

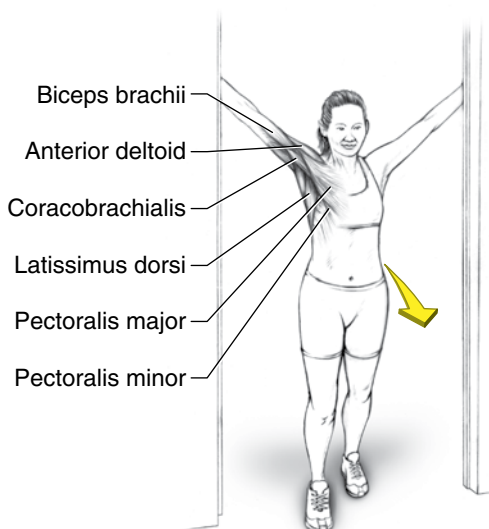
Technique

Stand upright while facing a doorway or corner.

Place feet shoulder-width apart with one foot slightly in front of the other.

With straight arms, raise arms high above the head, and place the palms on the walls or doorframe.

Lean the entire body forward.

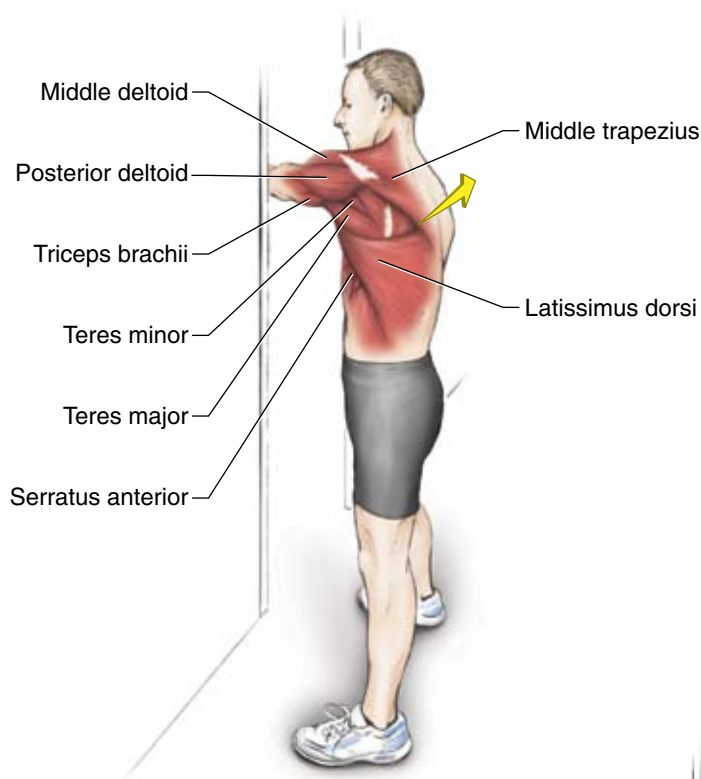


Muscles Stretched

Most-stretched muscles: Pectoralis major, anterior deltoid, coracobrachialis, biceps brachii, pectoralis minor.

Lesser-stretched muscles: Latissimus dorsi, lower trapezius, subclavius.

Shoulder Extensor, Adductor, and Retractor Stretch



Technique

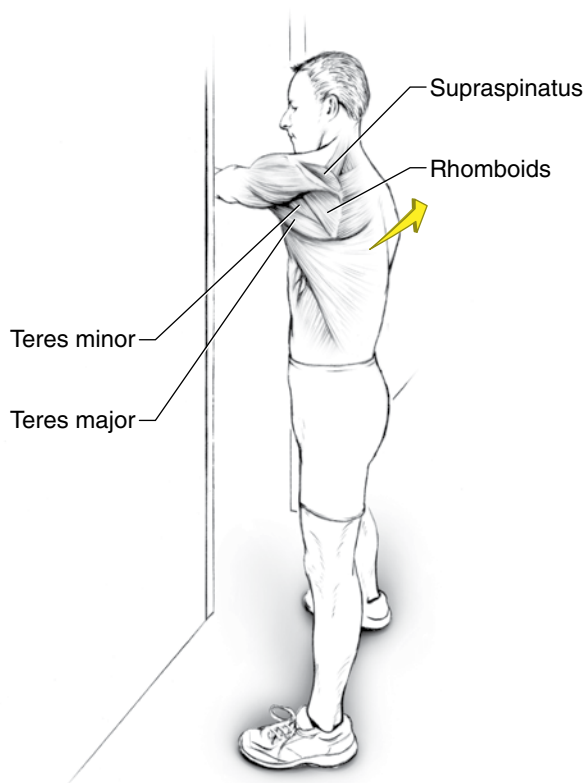
Stand upright inside a doorway while facing a doorjamb with the doorjamb in line with the right shoulder.

Place feet shoulder-width apart with the toes pointing straight forward.

Bring the left arm across the body toward the right shoulder.

Pointing the thumb down, grab hold of the doorjamb at shoulder level.

Rotate the trunk inward until you feel a stretch in the posterior left shoulder.



Muscles Stretched

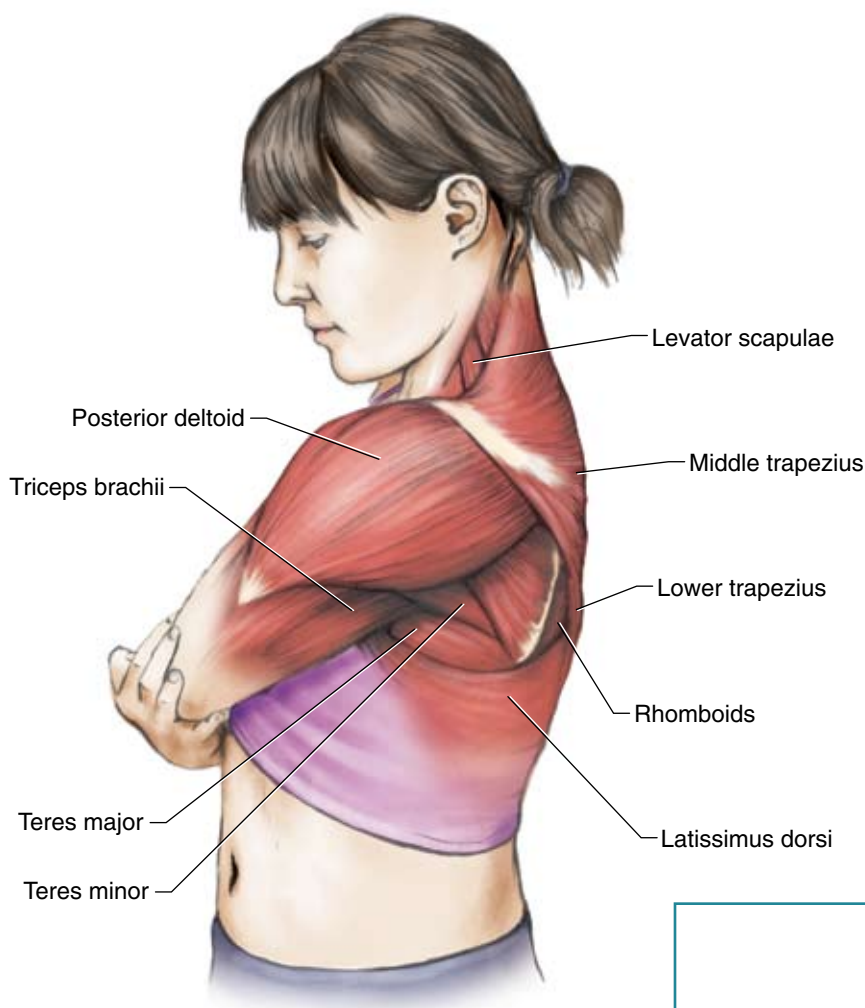
Most-stretched muscles: Left posterior and middle deltoid, left latissimus dorsi, left triceps brachii, left middle trapezius, left rhomboids.

Lesser-stretched muscles: Left teres major, left teres minor, left supraspinatus, left serratus anterior.

Commentary

To get the maximum benefit of this stretch, you should keep the elbow locked. Over time, as the muscles become more flexible, to keep the elbow locked you will need to grasp the doorframe above the level of the shoulder. Raising the height of the hand does not diminish the major benefits of this stretch. However, as the hand gets higher above shoulder level, the stretch on the rhomboids decreases while the stretch on the serratus anterior increases.

Shoulder Adductor, Protractor, and Elevator Stretch



Technique

Stand upright with the feet shoulder-width apart.

Bring the left arm across the front of the body, with the left hand near the right hip.

With the right hand, grab hold of the left elbow.

With the right hand, try to pull the left elbow down and around the right side of the body.



Muscles Stretched

Most-stretched muscles: Left posterior deltoid, left latissimus dorsi, left triceps brachii, left lower middle trapezius.

Lesser-stretched muscles: Left teres major, left teres minor, left supraspinatus, left levator scapulae, left rhomboids.

Commentary

To maximize the stretch, do not raise the shoulder or bend at the waist. If it is not possible to bring the hand toward the hip, try to come as close as possible. As long as the arm is below the shoulders, the stretch will be effective for the stated muscles.

VARIATION

Shoulder Adductor, Elevator, and Protractor Stretch

Bringing the arm above the shoulder changes the emphasis of the stretch to the elevators and protractors.

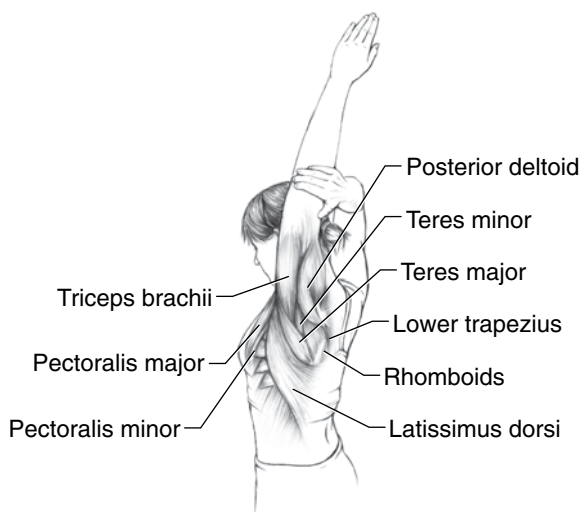
Technique

Stand upright with the feet shoulder-width apart.

Raise the left hand high above the head, and bring the left arm up against the left side of the head.

With the right hand, grab hold of the left elbow.

With the right hand, try to pull the left elbow behind the head, past the left ear.

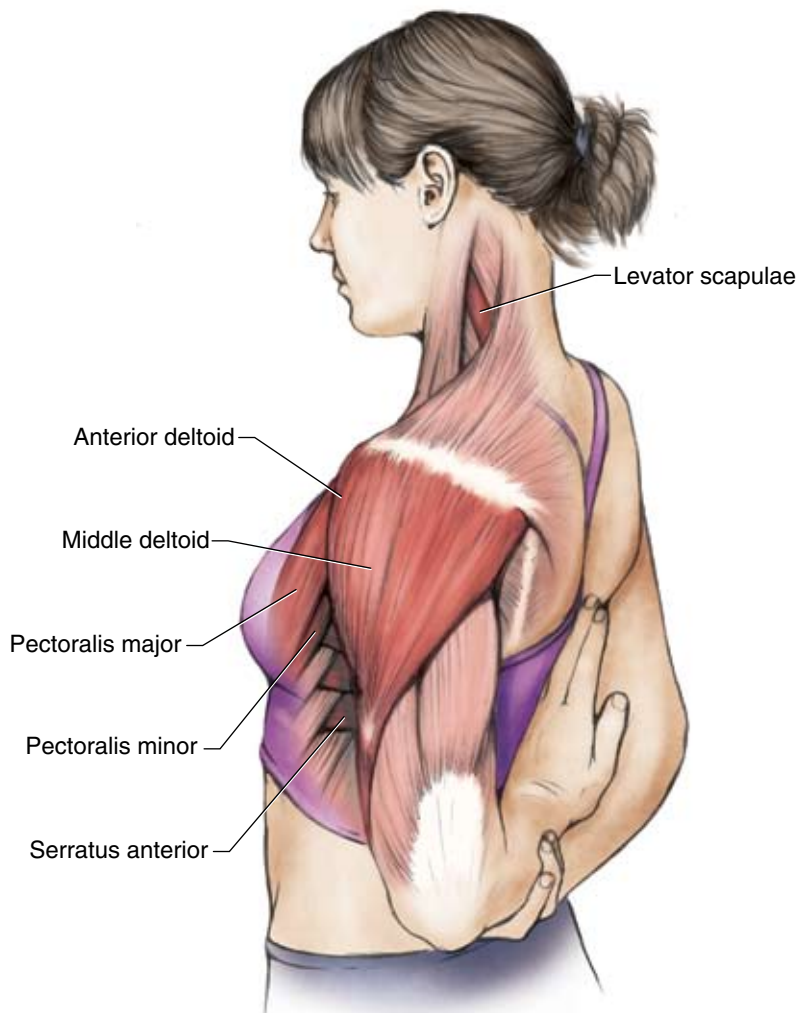


Muscles Stretched

Most-stretched muscles: Left posterior deltoid, left latissimus dorsi, left triceps brachii, left lower trapezius, left serratus anterior.

Lesser-stretched muscles: Left teres major, left teres minor, left supraspinatus, left rhomboids, left pectoralis minor.

One-Arm Shoulder Flexor Stretch



Technique

Stand (or sit on a backless chair) upright with the left arm behind the back and the elbow bent at about 90 degrees.

Place feet shoulder-width apart with the toes pointing forward.

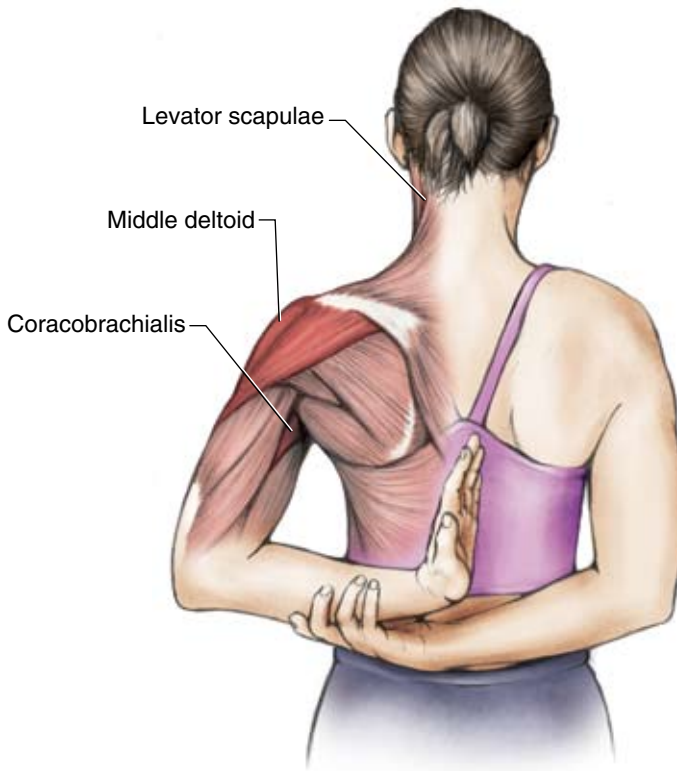
Grasp the left elbow with the right hand.

Pull the left arm across the back and up toward the right shoulder.

Muscles Stretched

Most-stretched muscles: Left pectoralis major, left anterior deltoid, and middle deltoid.

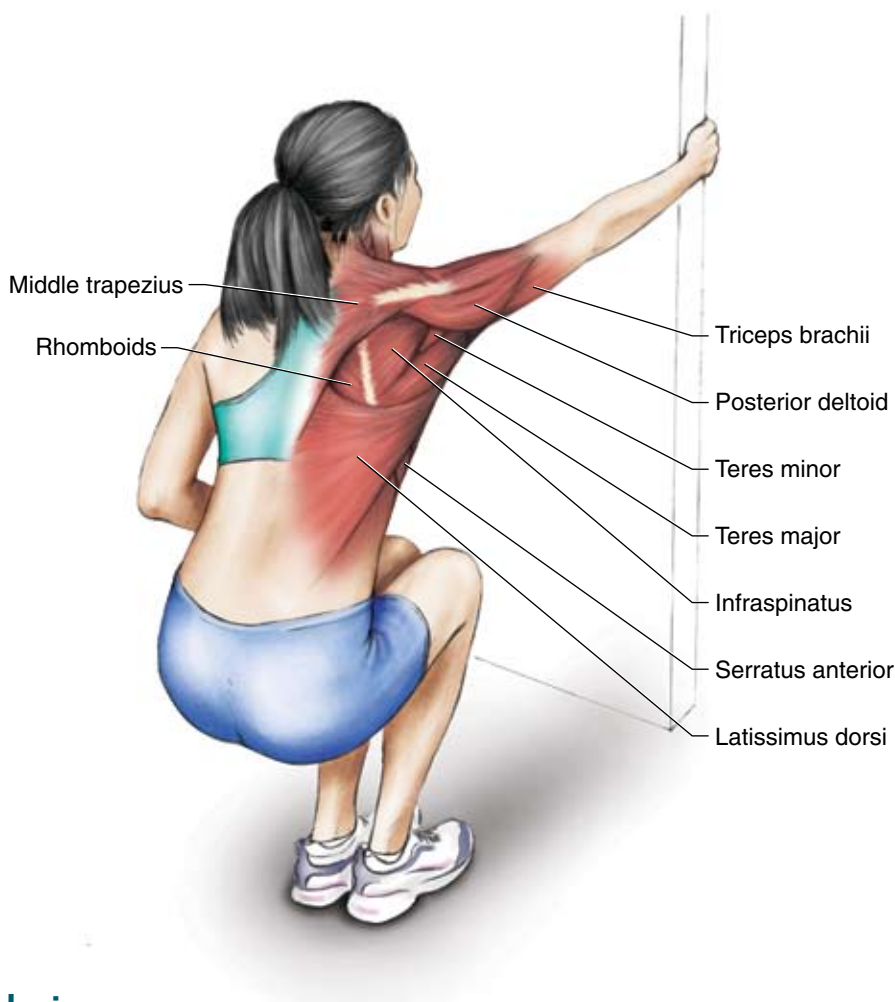
Lesser-stretched muscles: Left levator scapulae, left pectoralis minor, left supraspinatus, left serratus anterior, left coracobrachialis.



Commentary

If you cannot reach the elbow, then grasp the wrist. When pulling on the wrist, it is easy to pull the arm across the back, but remember that the best effect comes only from pulling upward as well as across. Also, keep the elbow locked at a near-90-degree angle. Changing the alignment of the back will also influence the magnitude of the stretch. If you cannot keep the back straight, arching the back is preferable to bending at the waist. Just be careful; it is easy to lose balance when doing this stretch while both arching the back and standing up.

Shoulder Adductor and Extensor Stretch



Technique

Stand in a squatting position while facing a doorway with the right shoulder lined up with the left side of the doorjamb.

Stick the right arm through the doorway. Grab the inside of the doorjamb at shoulder level with the right hand.

While keeping the right arm straight and the feet firmly planted, lower the buttocks toward the floor.

Muscles Stretched

Most-stretched muscles: Right posterior deltoid, right middle trapezius, right triceps brachii, right teres major, right rhomboids, right infraspinatus.

Lesser-stretched muscles: Right latissimus dorsi, right teres minor, right supraspinatus, right serratus anterior.

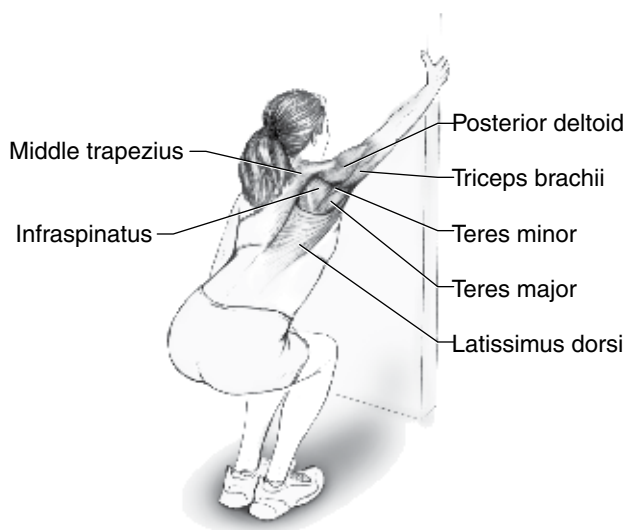
Commentary

A lower squat yields a greater stretch, but be careful not to squat so low that you feel pain in the legs or knees. To reduce strain on the knees, change where you grab the doorjamb. Changing the position of the grasp, however, influences the amount of stretch placed on the various muscles (see variation). Regardless of where you grasp, keep the back straight or arched. Do not bend forward at the waist. To get an even greater stretch, inwardly rotate the trunk.

VARIATION

Shoulder Adductor and Extensor Stretch Variation

Changing the hand position on the doorjamb changes the muscles that you stretch.



Technique

Stand in a squatting position while facing a doorway; line up the right shoulder with the left side of the doorjamb.

Stick the right arm through the doorway. With the right hand, grab the inside of the doorjamb above head level.

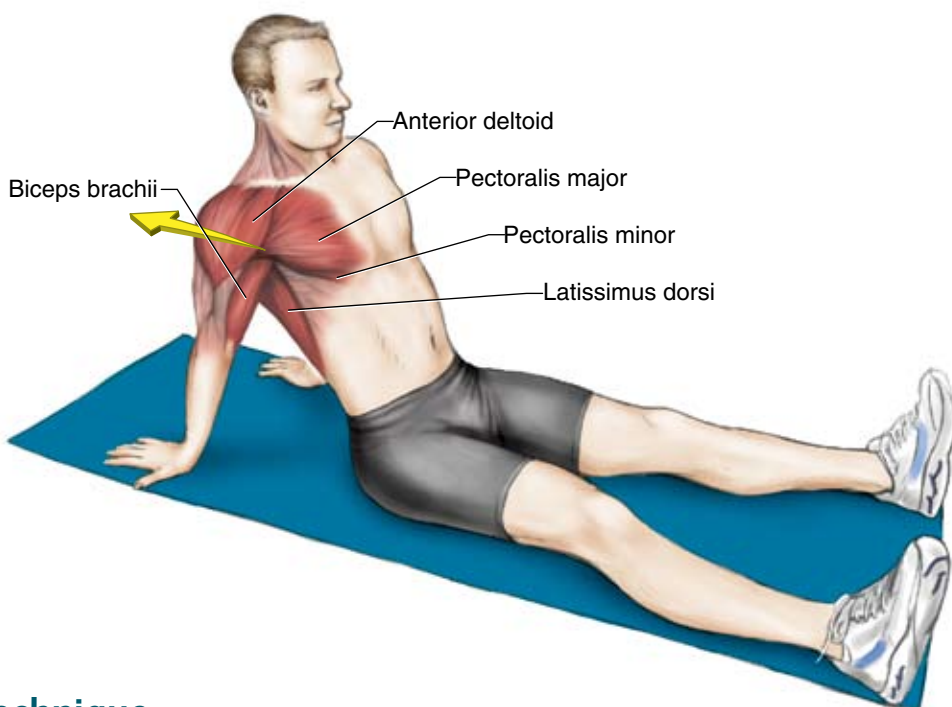
While keeping the right arm straight and the feet firmly planted, lower the buttocks toward the floor.

Muscles Stretched

Most-stretched muscles: Right posterior deltoid, right latissimus dorsi, right triceps brachii, right teres major, right infraspinatus.

Lesser-stretched muscles: Right teres minor, right supraspinatus, right middle trapezius.

Seated Shoulder Flexor Depressor Retractor Stretch



Technique

Sit on the floor with the legs straight.

While keeping the arms straight, place the palms (with the fingers pointed backward) on the floor about one foot (30 cm) behind the hips.

While keeping the arms straight, lean backward toward the floor.

Muscles Stretched

Most-stretched muscles: Pectoralis major, anterior deltoid, coracobrachialis, biceps brachii, pectoralis minor.

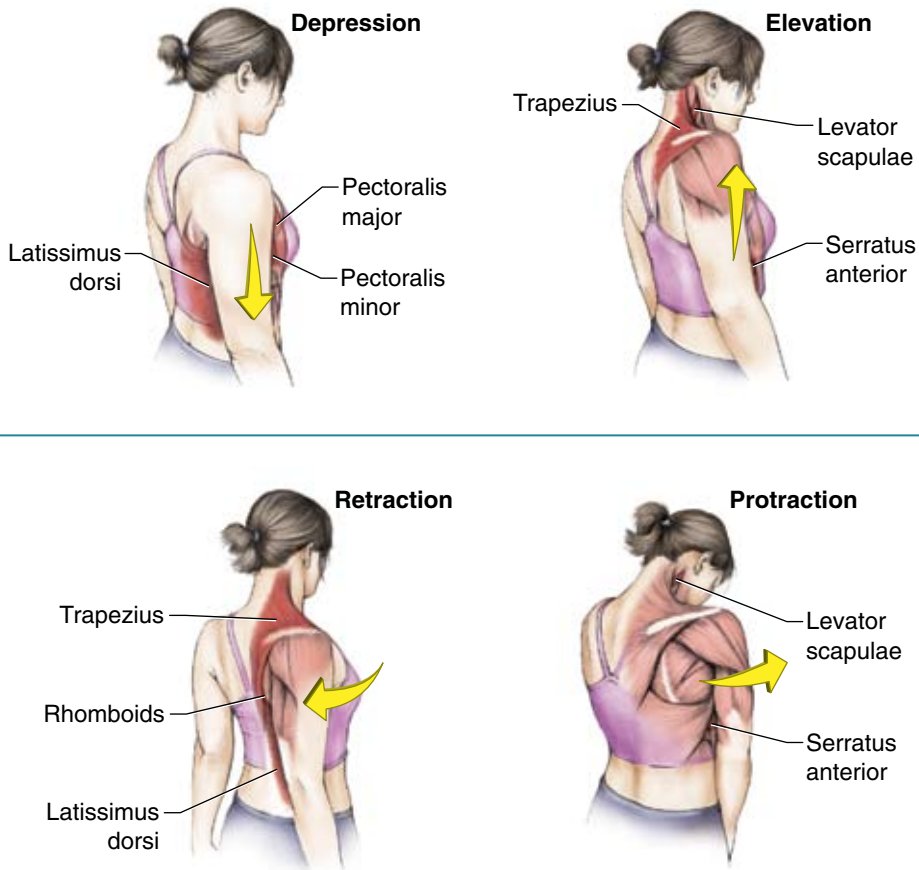
Lesser-stretched muscles: Latissimus dorsi, lower trapezius, subclavius, rhomboids.

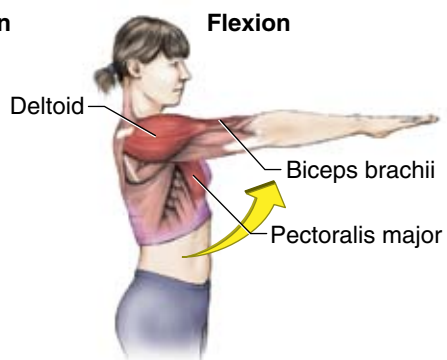
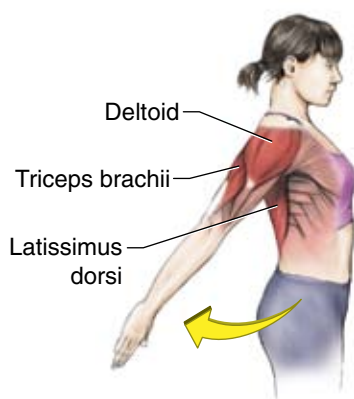
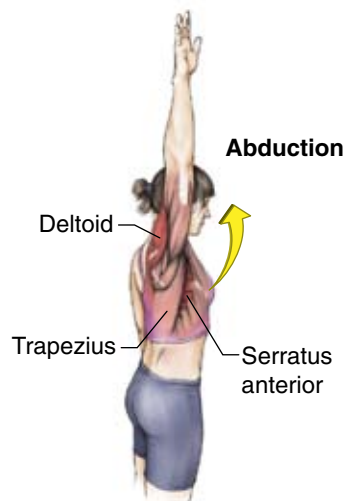
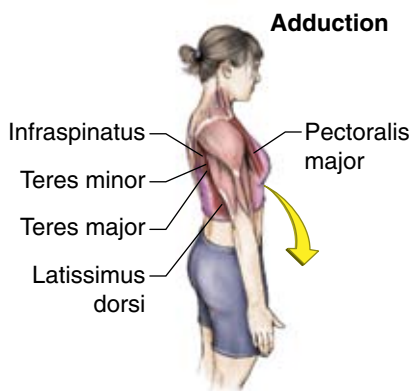
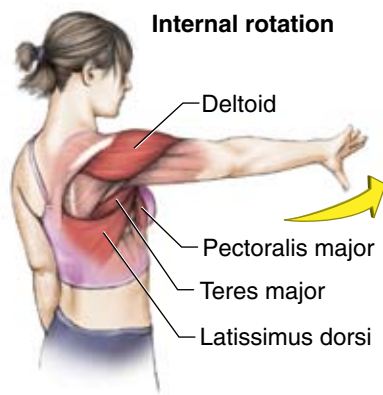
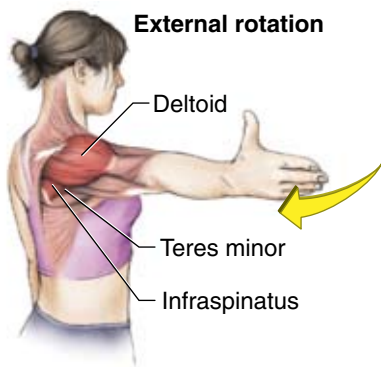
Commentary

To maximize the stretch, keep the arms straight. If it is difficult to refrain from bending the arms, place the hands closer to the hips. Moving the hands farther from the hips can increase the stretch. To keep the body from sliding along the floor, you may need to brace the soles of the feet against a wall. Sitting on a mat with the hands placed on a hard surface will increase the stretch as well as add comfort.

Shoulder, Back, and Chest Muscle Movements

The stretches in this chapter are excellent overall stretches; however, not all of these stretches may be completely suited to each person's needs. The muscles involved in the various shoulder and upper chest and back movements appear in the following table. To stretch specific muscles, the stretch must involve one or more movements in the opposite direction of the desired muscle's movements. For example, if you want to stretch the serratus anterior, you could perform a movement that involves shoulder depression, shoulder retraction, and shoulder adduction. When a muscle has a high level of stiffness, you should use very few simultaneous opposite movements (for example, to stretch a very tight pectoralis major, you would start by doing shoulder extension and external rotation). As a muscle becomes loose, you can incorporate more simultaneous opposite movements.





Muscle	Elevation	Depression	Protraction	Retraction	External rotation	Internal rotation	Abduction	Adduction	Flexion	Extension
Biceps brachii									✓	
Coracobrachialis								✓	✓	
Deltoid					✓	✓	✓		✓	✓
Infraspinatus					✓			✓		
Latissimus dorsi		✓		✓		✓		✓		✓
Levator scapulae	✓		✓							
Pectoralis major		✓				✓		✓	✓	
Pectoralis minor		✓	✓							
Rhomboids				✓						
Serratus anterior	✓		✓				✓			
Subclavius		✓								
Subscapularis						✓				
Supraspinatus							✓			
Teres major						✓		✓		✓
Teres minor					✓			✓		✓
Trapezius	✓			✓			✓			
Triceps brachii										✓